

42.3 Piecewise polynomials

Changing the basis left the second complaint untouched: polynomials are global, and a degree-16 fit spends the same sixteen degrees of freedom everywhere, so a feature at one end is paid for by an oscillation at the other. The remedy is a different space of functions: cut the range at a few points, fit a low-degree polynomial on each piece, and join the pieces smoothly. Nothing oscillates, and the result is still linear in its coefficients.

42.3.1 Knots and pieces

Definition 42.3.1. Piecewise polynomials and continuity constraints

Fix an interval $[a, b]$ and **knots** $a = \kappa_0 < \kappa_1 < \cdots < \kappa_K < \kappa_{K+1} = b$, the K inner ones being the **interior knots**. A function $f : [a, b] \rightarrow \mathbb{R}$ is a **piecewise polynomial of degree d** if its restriction to each half-open piece $[\kappa_k, \kappa_{k+1})$, $k = 0, \dots, K-1$, and to $[\kappa_K, b]$ agrees with a polynomial of degree at most d ; the value at a knot is the limit from the right, and nothing ties the pieces together. These form a vector space $\mathcal{PP}_d(\kappa)$ of dimension $(K+1)(d+1)$.

For $-1 \leq m \leq d-1$, f satisfies the **continuity constraints of order m** at κ_k if

$$f^{(r)}(\kappa_k^-) = f^{(r)}(\kappa_k^+), \quad r = 0, 1, \dots, m,$$

the case $m = -1$ imposing nothing. The **spline space $\mathcal{S}_d(\kappa)$** is the set of piecewise polynomials of degree d satisfying the constraints of order $d-1$ at every interior knot; its members are the **splines of degree d with knots κ** .

The constraints are linear in the coefficients: writing the unrestricted piecewise polynomial as $\sum_k \sum_{j \leq d} c_{kj} (t - \kappa_k)^j \mathbf{1}\{t \in [\kappa_k, \kappa_{k+1})\}$, the requirement $f^{(r)}(\kappa_k^-) = f^{(r)}(\kappa_k^+)$ reads

$$\sum_{j \geq r} \frac{j!}{(j-r)!} c_{k-1,j} (\kappa_k - \kappa_{k-1})^{j-r} = r! c_{kr},$$

one equation for each r and each interior knot. Fitting under them is restricted least squares (Proposition 11.3.3), and comparing two levels of smoothness is a general linear hypothesis (Theorem 11.3.2). Counting equations gives a candidate dimension, $(K+1)(d+1)$ coefficients minus $(m+1)K$ constraints, which for $m = d-1$ is

$$(K+1)(d+1) - dK = d+1 + K.$$

Only a candidate, since it assumes the constraints independent: the next theorem proves they are, by exhibiting a basis of exactly that size.

42.3.2 The spline space and the truncated power basis

Write $(v)_+ = \max(v, 0)$, and for $d \geq 1$ let

$$(t - \kappa)_+^d = \begin{cases} (t - \kappa)^d, & t \geq \kappa, \\ 0, & t < \kappa, \end{cases}$$

the **truncated power function** of degree d with knot κ . For $d = 0$ the same formula with the convention $0^0 = 1$ gives the indicator of $[\kappa, \infty)$.

Theorem 42.3.1. Dimension of the spline space and the truncated power basis

Let $d \geq 0$ and let $\kappa_1 < \dots < \kappa_K$ be interior knots of $[a, b]$.

(a) $\mathcal{S}_d(\kappa)$ is a vector space of dimension $d + 1 + K$, and

$$1, t, \dots, t^d, (t - \kappa_1)_+^d, \dots, (t - \kappa_K)_+^d \quad (42.3.1)$$

is a basis of it.

(b) More generally, if the constraints imposed at κ_k are of order m_k with $-1 \leq m_k \leq d-1$, the resulting space has dimension $d + 1 + \sum_{k=1}^K (d - m_k)$, with basis the powers $1, \dots, t^d$ together with $(t - \kappa_k)_+^r$ for $r = m_k + 1, \dots, d$ and $k = 1, \dots, K$.

Proof. It is enough to prove (b); part (a) is the case $m_k = d - 1$ for every k , for which the count is $d + 1 + K$.

The space. Sums and scalar multiples of piecewise polynomials of degree d are piecewise polynomials of degree d , and the constraints are linear equations, so the set is a subspace of the space of functions on $[a, b]$.

The listed functions belong to it. Each power t^j is a single polynomial, hence in every such space. Fix k and $r > m_k$. The function $g(t) = (t - \kappa_k)_+^r$ is zero on $[a, \kappa_k)$ and $(t - \kappa_k)^r$ on $[\kappa_k, b]$, a piecewise polynomial of degree $r \leq d$ whose only possible breakpoint is κ_k . For $s \leq r - 1$ its s th derivative at κ_k is $\{r!/(r-s)!\}(t - \kappa_k)^{r-s}$ there, which is zero, and zero from the left; so g satisfies the constraints of order $r - 1$, in particular those of order $m_k \leq r - 1$. At the other knots g is a single polynomial and satisfies any constraint.

Linear independence. Suppose

$$q(t) + \sum_{k=1}^K \sum_{r > m_k} c_{kr} (t - \kappa_k)_+^r = 0 \quad \text{for all } t \in [a, b],$$

with $q \in \mathcal{P}_d$. On $[a, \kappa_1)$ every truncated power vanishes, so q vanishes on an interval and hence $q \equiv 0$. Suppose inductively that $c_{lr} = 0$ for every $l < k$. On $[\kappa_k, \kappa_{k+1})$ the displayed sum reduces to $\sum_{r > m_k} c_{kr} (t - \kappa_k)^r$, a polynomial vanishing on an interval, so all its coefficients are zero. The induction gives $c_{kr} = 0$ throughout.

Spanning. Let f belong to the space. On $[\kappa_0, \kappa_1)$ it agrees with a polynomial $q \in \mathcal{P}_d$; set $f_1 = f - q$, again in the space and vanishing there. Suppose inductively that f_k is in the space and vanishes on $[a, \kappa_k)$, and on the next piece agrees with $r_k \in \mathcal{P}_d$. Expanding r_k in powers of $t - \kappa_k$,

$$r_k(t) = \sum_{s=0}^d \frac{r_k^{(s)}(\kappa_k)}{s!} (t - \kappa_k)^s,$$

and for $s \leq m_k$ the constraint at κ_k forces $r_k^{(s)}(\kappa_k) = f_k^{(s)}(\kappa_k^-) = 0$, because f_k is identically zero to the left. Hence $r_k(t) = \sum_{s > m_k} c_{ks} (t - \kappa_k)^s$ for some coefficients c_{ks} , and $f_{k+1} = f_k - \sum_{s > m_k} c_{ks} (t - \kappa_k)_+^s$ is in the space and vanishes on $[a, \kappa_{k+1})$: to the left of κ_k both terms vanish, and between κ_k and κ_{k+1} they cancel. After K steps f_{K+1} vanishes on all of $[a, b]$, which exhibits f as a combination of the listed functions.

Dimension. The listed functions are $d + 1 + \sum_k (d - m_k)$ in number, and they are a basis. ■

Remark. This settles the counting question: $(K + 1)(d + 1)$ minus $d + 1 + \sum_k (d - m_k)$ is $\sum_k (m_k + 1)$, exactly the number of constraint equations. The constraints are therefore independent, and that a free piecewise polynomial is in fact a spline is testable with $\sum_k (m_k + 1)$ degrees of freedom (Definition 11.3.1).

42.3.3 Three familiar special cases

Degree 0. $\mathcal{S}_0(\kappa)$ is the space of step functions, of dimension $K + 1$, and fitting it is the one-way layout with the intervals as groups (Example (The one-way layout)): the fit is *exactly local*, with variance σ^2/n_k on interval k , at the price of the discontinuity and a bias of order the interval width.

Degree 1. $\mathcal{S}_1(\kappa)$, of dimension $K + 2$, is the space of continuous broken lines, with basis $1, t, (t - \kappa_1)_+, \dots, (t - \kappa_K)_+$; the coefficient of $(t - \kappa_k)_+$ is the *change* in slope there, and the two-piece case is the classical two-phase regression.

Degree 3. $\mathcal{S}_3(\kappa)$ is the space of **cubic splines**, of dimension $K + 4$: value, slope and curvature continuous, only the third derivative jumping. Cubic is the default, the lowest degree whose joins are invisible and whose second derivative a smoothness penalty can control.

Warning. A knot estimated from the data takes the model out of this chapter: $(t - \kappa)_+^d$ is not linear in κ , the criterion is not even differentiable in it for $d = 0$ or 1 , and the usual asymptotics can fail. This **segmented** or **changepoint** regression has a literature of its own — Hudson (1966), Feder (1975), Seber and Wild (1989, ch. 9) — which this book does not develop. The honest options are to fix the knot from outside the data, to fit many fixed knots and let a penalty decide (Chapter 43), or to pay for the selection (Proposition 29.5.1).

42.3.4 Climbing the smoothness ladder

Example. Four fits with the same knots

Take the Nile series again, $n = 100$ annual flows, with cubic pieces and interior knots at 1899 — where Cobb (1978) and the analysis of Example (A trend in the Nile) both locate a change — and 1935. Fitting the three cubics separately uses 12 parameters and leaves a residual sum of squares of 1394072. Continuity at both knots removes two parameters and raises it to 1561452; continuity of the first derivative removes two more (1613581); and the full cubic spline, with 6 parameters as Theorem 42.3.1 predicts, has 1781797. Figure 42.3.1 shows the four fits.

Each step down the ladder is a testable linear hypothesis, and against the unrestricted fit all three are rejected: $F = 5.28$ on $(2, 88)$ degrees of freedom for continuity alone ($p = 0.0068$), $F = 3.46$ on $(4, 88)$ ($p = 0.0112$) for a continuous first derivative, and $F = 4.08$ on $(6, 88)$ ($p = 0.0012$) for the cubic spline. The Nile really does have a discontinuity, and a spline is

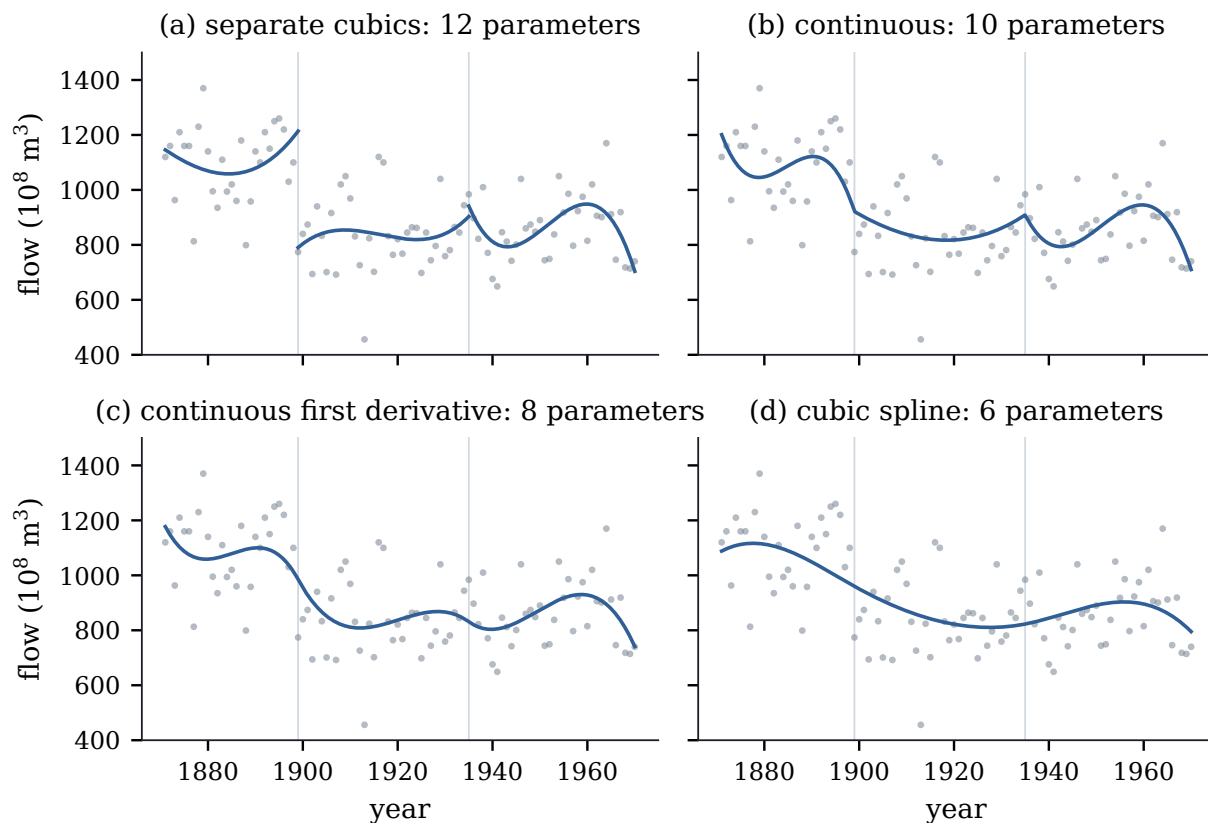


Figure 42.3.1: The Nile flows with cubic pieces cut at 1899 and 1935, under four levels of continuity; each constraint removes two parameters.

the wrong space for it.

The right model is the simplest on the ladder: a step function with a single knot at 1899, whose group means are 1097.7 and 850.0 in units of 10^8 m^3 , differing by -247.8 with standard error 28.4 and $t = -8.71$. Its residual sum of squares, 1597457, beats the six-parameter spline and the cubic polynomial's 1909955. Two parameters in the right place beat six in the wrong space.

```
import numpy as np
import statsmodels.api as sm

nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy().astype(float)
y = nile["volume"].to_numpy()
n = len(y)
knots = np.array([1899.0, 1935.0])
d = 3
u = lambda s: 2 * (s - year.min()) / (year.max() - year.min()) - 1

def separate(s):
    """One free cubic per interval: (K+1)(d+1) columns, no constraints at the knots.
    """
    piece = np.digitize(s, knots)
    return np.column_stack([np.where(piece == k, u(s) ** j, 0.0)
                            for k in range(len(knots) + 1) for j in range(d + 1)])
```

```

def truncated(s, m):
    """Continuity of the function and of its first m derivatives at every knot."""
    cols = [u(s) ** j for j in range(d + 1)]
    for k in knots:
        cols += [np.clip(u(s) - u(k), 0.0, None) ** r for r in range(m + 1, d + 1)]
    return np.column_stack(cols)

def fit(X):
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    resid = y - X @ beta
    return np.linalg.matrix_rank(X), resid @ resid, X @ beta

for name, X in [("separate cubics", separate(year)), ("continuous", truncated(year,
    0)),
                ("continuous derivative", truncated(year, 1)), ("cubic spline",
    truncated(year, 2))]:
    p, rss, _ = fit(X)
    print(f"{name:24s} p = {p:2d}    residual sum of squares {rss:10.0f}")

```

[Run this code online](#)

```

from scipy import stats

p_full, rss_full, _ = fit(separate(year))
for m, label in ((0, "continuity"), (1, "first derivative"), (2, "second derivative"
    )):
    p_m, rss_m, _ = fit(truncated(year, m))
    q = p_full - p_m
    F = ((rss_m - rss_full) / q) / (rss_full / (n - p_full))
    print(f"up to the {label:17s}: q = {q}, F = {F:5.2f}, p-value {1 - stats.f.cdf(F
    , q, n - p_full):.4f}")

```

[Run this code online](#)

Warning. The knot at 1899 was not chosen blind: Cobb's analysis and the large quartic component in Example (A trend in the Nile) both point to the end of the nineteenth century, and both used this series. The t statistic -8.71 therefore overstates the evidence for a break of unknown date, whose null distribution is that of the largest of ninety-nine statistics, one for each year one could split at.

Idea. Degrees of freedom are the currency, and a spline spends them differently. Each of a polynomial's $d + 1$ parameters affects the curve everywhere; a spline has $d + 1 + K$, and the K extra act only to the right of their knots, adjusting the d th derivative. Raising K at fixed low d is a local purchase; raising d is not.

Remark (Natural boundary conditions). Outside $[\kappa_1, \kappa_K]$ a cubic spline is still a cubic, and cubics diverge. **Natural** cubic splines add $f''(a) = f'''(a) = f''(b) = f'''(b) = 0$, forcing f to be linear beyond the outer knots and reducing the dimension to K ([Exercise \(B1\)](#)) — a

decision that the curve should flatten where the data run out, not a discovery about them. It becomes more than a convention in [Chapter 43](#), where the natural cubic spline exactly minimizes a penalized criterion (Theorem 43.4.2).

42.3.5 Exercises

42.3.5.1 A. Check your understanding

Exercise. A1

How many parameters has a quadratic spline with 7 interior knots? A cubic spline with 7, one of them a double knot (only C^1 there)? A piecewise constant function with 7?

Solution. By [Theorem 42.3.1](#), $2 + 1 + 7 = 10$, then $3 + 1 + 7 + 1 = 12$ (the double knot contributes $d - m = 3 - 1 = 2$ instead of 1), and $0 + 1 + 7 = 8$.

Exercise. A2

In the broken-line model $f(t) = \beta_0 + \beta_1 t + \sum_k \theta_k (t - \kappa_k)_+$, give the slope of f on (κ_j, κ_{j+1}) , the hypothesis that the line is unbent at κ_j , and the degrees of freedom of the test that it is a single straight line.

Exercise. A3

Verify that for the piecewise constant fit, changing y_i by Δ moves the fitted curve by Δ/n_k on i 's own interval and not at all elsewhere. Contrast with Proposition 42.1.1(d).

42.3.5.2 B. Practice

Exercise. B1

Show that the natural cubic splines with interior knots $\kappa_1 < \dots < \kappa_K$ ($K \geq 2$) form a space of dimension K , by writing the four boundary conditions in the truncated power basis and checking that they are independent.

Solution. Write $f(t) = \sum_{j \leq 3} \beta_j t^j + \sum_k \theta_k (t - \kappa_k)_+^3$, with $K + 4$ coefficients. Linearity on $[a, \kappa_1]$ means $\beta_2 = \beta_3 = 0$; linearity on $[\kappa_K, b]$ makes the cubic and quadratic coefficients of the last piece vanish, which after expanding $(t - \kappa_k)^3$ reads $\sum_k \theta_k = 0$ and $\sum_k \theta_k \kappa_k = 0$. The four equations involve different coefficients and are independent, so the dimension is K .

Exercise. B2

Two straight lines meet at a known γ . Write the model in the basis $1, t, (t - \gamma)_+$, give the

least squares estimate of the common value at γ and its variance, and the F statistic for equal slopes.

Solution. With $E(Y) = \beta_0 + \beta_1 t + \theta(t - \gamma)_+$, the value at γ is the estimable function $\beta_0 + \beta_1 \gamma$, estimated by $\hat{\beta}_0 + \gamma \hat{\beta}_1$ with variance $\sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$, $\mathbf{a} = (1, \gamma, 0)^\top$. Equal slopes is $\theta = 0$, tested by the square of the t statistic for $\hat{\theta}$ against $F(1, n - 3)$ (Theorem 11.6.1).

Exercise. B3

Let $f \in \mathcal{S}_d(\kappa)$ have truncated power coefficients θ_k . Show that $f^{(d)}$ is a step function jumping by $d! \theta_k$ at κ_k , and that $\theta_k = 0$ exactly when f is a single polynomial across κ_k .

Solution. Differentiating Eq. 42.3.1 d times kills every power below t^d and turns $(t - \kappa_k)_+^d$ into $d!$ times the indicator of $[\kappa_k, \infty)$, so $f^{(d)} = d! \beta_d + d! \sum_k \theta_k \mathbf{1}\{t \geq \kappa_k\}$. If $\theta_k = 0$ the pieces on either side of κ_k have the same d th derivative and agree to order $d - 1$ there, hence are the same polynomial; the converse is immediate.

42.3.5.3 C. Going deeper

Exercise. C1

Show that for $f \in C^{d+1}[a, b]$,

$$f(t) = \sum_{j=0}^d \frac{f^{(j)}(a)}{j!} (t - a)^j + \frac{1}{d!} \int_a^b f^{(d+1)}(s) (t - s)_+^d ds.$$

Interpret it as saying that the truncated power functions are a continuous basis for smooth functions, of which Eq. 42.3.1 keeps K members.

Exercise. C2

Let κ and $\kappa + \epsilon$ be two knots of a cubic spline. Show that as $\epsilon \downarrow 0$ the angle between $(t - \kappa)_+^3$ and $(t - \kappa - \epsilon)_+^3$ tends to zero, while the normalized difference $\{(t - \kappa)_+^3 - (t - \kappa - \epsilon)_+^3\} / (3\epsilon)$ tends to $(t - \kappa)_+^2$. Explain how this both shows that a double knot lowers the smoothness by one order and warns against placing knots close together in the truncated power basis.

Solution. For $t \geq \kappa + \epsilon$, $\{(t - \kappa)^3 - (t - \kappa - \epsilon)^3\} / (3\epsilon) \rightarrow (t - \kappa)^2$ by the definition of the derivative; for $t \leq \kappa$ both terms vanish, and the middle interval, of width ϵ , contributes nothing. The limit is $(t - \kappa)_+^2$, the basis function Theorem 42.3.1(b) adds at a knot of multiplicity two, where only C^1 continuity is required. Two nearly coincident columns whose difference carries the information is what makes a design ill-conditioned (Definition 1.8.1), and it inflates the coefficients' variances in the sense of Definition 26.2.1: the fit is fine, the coefficients are not.